Anaconda v4.2.2 - Localized Body Compression

STAGE 2 IMPLEMENTATION REPORT | base file: anaconda-wordpress-snippet-v4.2.1.html | deliverable: anaconda-wordpress-snippet-v4.2.2.html

1. Findings from the v4.2.1 inspection

Before changing anything, the existing body pipeline was traced end to end. The visual tube is rebuilt every frame in **AnacondaRig._updateBody()** around an 80-ring physics spine; each ring gets a cross-section whose belly is statically flattened (HV = 0.94 vertical squash, BF = 0.90 belly flatten) so the body rests on the ground. Two contact systems already existed: (a) **self-contact** is fully solved since v4.1/v4.2 - the strongest overlap per ring is measured in 3D on the visual chain (_ctA/_ctN* fields), temporally smoothed, blurred along the spine (5-tap binomial), and consumed by the cross-section as a directional "flesh squash" with lateral mush-out and a crease shadow; (b) **ground contact** had no dynamic response at all: every ring sits at a fixed resting height (y = 0.02 + lift + bob) and the belly flatten was the same anatomical constant everywhere, whether the section was resting, pushing, or lifted clear of the ground. Weight existed nowhere in the model.

Conclusion: body-against-body pressing already behaves correctly and is reused untouched; the missing half of the brief was localized, weight-dependent **ground** compression with smooth onset and recovery.

2. What changed (80 changed lines, all inside the skin-tube system)

A per-ring **ground-press field** was added, mirroring the existing overlap-field machinery. It is built only from signals that already exist - nothing is invented, and rings whose belly is measurably off the ground get zero compression:

```
gapY = this._R[i].y - 0.02 /* measured height above resting position */
contact = 1 - clamp(gapY / 0.055, 0, 1) /* lifted front third gets none */
wgt = RAD[i]^2 / RADMAX^2 /* local weight: mass per unit length */
pushW = max(0, sin(_waveD - i*waveK)) * speed01 /* gait push wave */
raw = contact * wgt * smoothstep(i/6) * (0.045*idleAmt + 0.024*pushW)
```

Change	Where	Detail
New field arrays	AnacondaRig constructor	_gPr (raw), _gSm (temporally smoothed), _gA2 (spine-blurred press depth), plus RADMAX normaliser.
Press field build + continuity pipeline	_updateBody(), after the overlap-field block	Raw press from the signals above; exponential smoothing in seconds (rise 9/s, fall 3.2/s), then the same 5-tap binomial spine blur the overlap field uses. Neck fade (rings 0-6, smoothstep) protects the skull-loft junction, which is built with the static cross-section. Hard cap 0.06 like every field in the rig.
Localized cross-section compression	_updateBody() ring loop	The static belly flatten deepens per ring: HVr = HV - gA2[i]; the ring centre is lowered by exactly the amount the deeper flatten pulls the bottom edge up (r x (BOT - 2.56 x gP)), so the flattened belly stays anchored on the ground line while the top sinks (3.36 x r x depth) and the lower flanks spread slightly through the existing girth-keeping ratio (vf2 = 1 + (1-vf) x 0.55). Volume compensation is inherited from the same ratio the static flatten uses - the body reads as pressed muscle, never inflated. Normals are analytic and follow the new offsets automatically.
Version + changelog	File header	v4.2.1 to v4.2.2 bump and a changelog block; nothing else touched.

3. How continuity, attachment and recovery are guaranteed

No boundaries, no popping. The raw press field is already smooth along the body (sine wave, scalar settle, smooth profile), and it then passes through the identical continuity pipeline as the v4.2 overlap field: exponential rise (about 0.11 s time constant) and slower decay (about 0.31 s), then a 5-tap binomial blur along the spine. Measured worst ring-to-ring step in the settled field: 0.006 of press depth - impossible to see as a segment boundary. Around the section, the deeper flatten fades linearly to zero at the widest line, exactly like the existing static belly flatten, so no

new edge is introduced. Recovery is exponential decay of the same field: when contact ends the section eases back (measured worst per-frame change 0.0023, i.e. about 0.14 per second) and never snaps.

Skin stays attached. The markings are one continuous texture-space skin (UV per vertex, fixed); compression only moves vertices in the tube geometry, so blotches and ocelli ride the flattened area exactly as before - verified in close-up A/B renders (figure delivered separately): pattern continuous across the pressed zone, at most an imperceptible sub-percent scale stretch, same as real skin.

Movement untouched. step(), the brain, gaits, steering, relax physics, head, tongue, lighting and camera are byte-identical to v4.2.1 (verified by diff; the field only reads existing state: _waveD, _gp, _speed, _R[i].y, idleAmt).

4. Validation results (headless browser, instrumented rig)

Required check	Result	Measured evidence
1. Normal movement without contact	PASS	Free crawl at speed 0.39: press max 0.0035, mean 0.0002 - only faint travelling push bands; no uniform squash; zero JS errors.
2. Body resting against the surface	PASS	At rest the press settles at 0.045 at ring 27 (deepest girth), 0.012 neck, 0.007 tail - weight profile works; belly bottom stays on the ground line (ON vs recovered bottom dY = 0.000008).
3. Section pressing against another section	PASS	Hairpin fold: existing overlap field reaches strength 1.0 with squash + crease shading; composed cleanly with ground press.
4. Skin patterns stay attached	PASS	Close-up A/B renders: blotches/ocelli continuous across the compressed zone; no swimming (UV-locked texture).
5. Body smooth and continuous	PASS	Max ring-to-ring press step 0.006; top sink 0.0176 measured for depth 0.0145 - matches the 3.36 x r x d model exactly; silhouette continuous in all renders including a x3.2 debug exaggeration.
6. Movement and behaviour unchanged	PASS	step()/brain/gaits byte-identical (diff); live runs error-free; boot test of the deliverable file: booted, animated, 0 errors.

Recovery timing: half of the press decays in about 0.28 s after the animal starts moving; full settle-back within a few seconds, monotonic and smooth.

5. Limitations and items not verified

(a) The scene has a flat ground plane and the body is always within a fraction of a radius of it, so "ground contact" here means the existing resting height, not a generic terrain solver - the field keys off the ring height above that resting position, which is exact for this architecture. (b) Body-on-body compression is the pre-existing v4.1/v4.2 system, reused as-is; its known behaviour is unchanged. (c) Magnitudes were tuned numerically in headless Chromium; low-end adaptive-quality tiers share the same field logic but were only smoke-tested. (d) No underwater, slope or obstacle geometry exists in the scene, so compression against such surfaces is out of scope and untested.

6. Files delivered

anaconda-wordpress-snippet-v4.2.2.html (drop-in replacement, self-contained, no new dependencies); **anaconda-compression-before-after.png** (labelled A/B panels: recovered vs pressed, debug exaggeration, body-on-body); this report.